

MediKiosk

AI-Powered Patient Case-Taking Software

Combined Problem Statement, Evidence Base & Solution Documentation

Field	Detail
Problem Statement ID	SIH26047
Title	Patient Case-Taking Software
Organisation	All India Institute of Ayurveda (AIIA), Ministry of Ayush
Category	Software
Theme	MedTech / BioTech / HealthTech
Team	Byte-Craftsman-Alpha
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Official Problem Statement (SIH26047)

This section reproduces the official Smart India Hackathon problem statement issued by the sponsoring organisation, exactly as released, for reference.

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Theme	MedTech / BioTech / HealthTech
YouTube / Video Link	NA
Dataset Link	NA

Background

1.1 The Clinical History-Taking Bottleneck in Indian Hospitals

History taking — the structured elicitation of a patient's presenting complaints, history of present illness, past medical and surgical history, drug and allergy history, family and personal history, and a review of systems — is the single most important diagnostic activity in clinical medicine. Classical teaching holds that a well-conducted history yields the correct diagnosis in **70–80% of cases**, even before examination or investigation. Yet in India's overburdened public hospital outpatient departments (OPDs), the time available for this critical interaction has collapsed to unsustainable levels.

India operates one of the most patient-dense healthcare systems in the world. Tertiary government hospitals and apex institutions routinely register **4,000–10,000 OPD patients per day**, with a doctor-to-patient consultation time frequently reported between **2 and 5 minutes** — among the shortest globally (study published in BMJ Open, 2017, across 67 countries placed India's average primary-care consultation at just over 2 minutes). Within this window, the physician must simultaneously elicit history, examine the patient, review prior records, formulate a diagnosis, counsel, and prescribe. The result is systematic under-elicitation of history, missed comorbidities, repeated questioning across visits, and diagnostic error.

AYUSH institutions face an additional layer of complexity. Ayurvedic history taking (Trividha, Ashtavidha, and Dashavidha Pariksha) requires detailed assessment of Prakriti (constitution), Vikriti (current imbalance), Agni (digestive capacity), Koshtha (bowel nature), Ahara-Vihara (diet and lifestyle), Nidana (causative factors), and Samprapti (pathogenesis) — a far more extensive history framework than allopathic intake.

Capturing this depth manually within OPD time constraints is effectively impossible, forcing practitioners to abbreviate the very assessment that defines personalized Ayurvedic care.

1.2 The Documentation and Records Fragmentation Problem

Compounding the time problem is the fragmentation of patient records. Patients in India typically carry physical paper prescriptions, laboratory reports, discharge summaries, and imaging films from multiple prior providers. During consultation, the physician must manually scan through these unstructured documents — often handwritten, in varying languages, and chronologically disordered — consuming a significant fraction of the already-scarce consultation time. There is no point-of-entry mechanism to digitize, structure, and chronologically organize a patient's prior medical documents before they reach the consultation room.

The Ayushman Bharat Digital Mission (ABDM) has established the national digital health infrastructure — ABHA (Ayushman Bharat Health Account) IDs, the Health Information Exchange, and FHIR-based interoperability standards. However, the 'first-mile' problem remains unsolved: there is no efficient, patient-facing software platform that captures structured history and digitizes documents into the ABDM ecosystem before the clinical encounter begins.

1.3 The Opportunity: AI-Powered Digital Clinical Intake Platform

Self-service kiosks have transformed high-throughput service industries — ATMs in banking, self-check-in terminals in aviation, and ordering kiosks in quick-service restaurants — by offloading structured data-entry tasks from human staff to the user, dramatically improving throughput and accuracy. In healthcare, patient check-in kiosks are now widespread in developed-country hospitals, but these are limited to administrative check-in. None perform deep, AI-driven, multimodal clinical history acquisition with medical document digitization.

The convergence of mature enabling technologies — robust automatic speech recognition (ASR) for Indian languages and accents (Bhashini / AI4Bharat models), large language models for conversational clinical history elicitation, high-accuracy OCR for handwritten and printed medical documents, and ABDM's FHIR interoperability — now makes it feasible to build an AI-powered clinical history software platform.

Description

2.1 The Problem in Precise Terms

There is no purpose-built, patient-facing software platform that enables patients to independently and comprehensively record their medical history — through both natural spoken conversation and guided touchscreen interaction — and simultaneously digitize their existing physical medical documents, generating a structured, physician-ready clinical history summary that integrates with the hospital information system and the ABDM ecosystem before the patient enters the consultation room.

2.2 Why Existing Solutions Fall Short

- **Existing hospital registration systems** (currently deployed in some Indian hospitals) capture only demographic and appointment data — name, age, department, token number. They do not elicit any clinical history or process medical documents.

- **Mobile health apps and tele-triage chatbots** require smartphone literacy, stable connectivity, and patient enrolment ahead of the visit — excluding the large elderly, rural, low-literacy, and first-visit patient populations who form the bulk of government hospital OPD load.
- **Manual nurse-led triage / history desks** are themselves human-resource-limited, do not scale to 5,000+ daily patients, and reintroduce the same time and transcription bottleneck the system is trying to eliminate.
- **Generic document scanners** digitize images but do not extract, structure, or chronologically organize clinical content, nor link it to a structured history or ABHA record.

2.3 Specific Challenges a Solution Must Overcome

- **Multilingual, multi-accent voice capture** in noisy hospital environments across Hindi, English, and major regional languages, for patients of varying literacy and digital comfort.
- **Accessibility for low-literacy and elderly users** through intuitive icon-driven UI, audio prompts, and conversational guidance — the software platform must be usable by a first-time, non-tech-savvy patient with zero training.
- **Accurate clinical history structuring** converting free-form patient narration into a standardized, physician-readable history (chief complaint, HPI, past history, drug/allergy, family, personal, review of systems) — and, for AYUSH settings, Dashavidha Pariksha parameters.
- **Reliable medical document digitization** OCR of handwritten and printed prescriptions, lab reports, and discharge summaries in multiple languages, with intelligent extraction of diagnoses, medications, and investigation values.
- **Privacy, consent, and data security** compliance with the Digital Personal Data Protection Act 2023 and ABDM consent framework — handling sensitive health data within a secure software environment.

Expected Solution (as released)

3.1 Solution Overview — “MediKiosk” AI Clinical History Software Platform

The proposed solution — tentatively designated MediKiosk — a software platform for an AI-powered clinical history software platform that allows any patient to record a comprehensive medical history through natural voice conversation and guided touchscreen interaction, scan and digitize their existing physical medical documents, and generate a structured, physician-ready clinical history summary that is pushed to the hospital information system (HIS) and linked to the patient's ABHA record — all completed before the consultation, with minimal staff assistance required.

3.2 Software Architecture (as specified in the PS)

Component	Specification & Rationale
Accessibility	Audio-guided mode for visually impaired; large-text/high-contrast mode; multilingual UI; sign-language video avatar (stretch goal)

Component	Specification & Rationale
Patient dashboard	Personal login id and password for patients to put data and store medical records

3.3 Software & AI Stack (Integrated) — as specified in the PS

Module A — Conversational Multimodal History Engine. A conversational AI engine that conducts a structured clinical history interview through both voice and touch. The patient speaks naturally in their preferred language; the engine asks intelligent follow-up questions (e.g., on stating 'chest pain', it probes onset, character, radiation, aggravating/relieving factors — the SOCRATES framework) and simultaneously offers touch-based multiple-choice options for patients who prefer tapping.

- **Adaptive questioning:** dynamically branches based on chief complaint and prior answers, mirroring a physician's clinical reasoning to elicit a complete HPI and review of systems.
- **Dual-mode input:** every question answerable by speaking OR tapping, ensuring usability across literacy and comfort levels.
- **AYUSH history mode:** for Ayurvedic OPDs, an extended interview capturing Dashavidha Pariksha (Prakriti, Vikriti, Sara, Samhanana, Pramana, Satmya, Sattva, Ahara Shakti, Vyayama Shakti, Vaya) and Ahara-Vihara assessment.
- **Red-flag detection:** AI flags emergency symptoms (e.g., acute chest pain with dyspnoea, stroke symptoms) and triggers immediate priority alert to triage staff rather than routine queueing.

Module B — Medical Document Digitization & Intelligence. An integrated scanning and document-AI pipeline that allows the patient to upload prior prescriptions, lab reports, and discharge summaries. The system performs high-accuracy OCR (printed and handwritten, multilingual), then extracts and structures clinical entities.

- **Intelligent extraction:** diagnoses, prescribed medications with dosages, investigation results with values and reference ranges, and procedure/surgery history.
- **Chronological organization:** automatically dates and orders documents into a coherent medical timeline for the physician.
- **Abnormal-value highlighting:** flags out-of-range lab values and potential drug interactions for physician attention.

Module C — Structured History Summary Generator. An AI summarization engine that synthesizes the conversational history and the digitized documents into a single, concise, physician-ready clinical summary in standard format — presented on the consultation screen the moment the patient enters the room.

- **Standard clinical format:** Chief complaint → HPI → Past medical/surgical → Drug & allergy → Family → Personal → ROS → Prior investigations summary.
- **Editable & verifiable:** physician retains full control — the summary is a draft to accept, amend, or reject, never an autonomous diagnosis.

- **Bilingual output:** patient-facing audio confirmation in local language; physician-facing summary in English/Hindi.

Module D — Consent, Privacy & ABDM Integration. A robust consent and security layer compliant with the Digital Personal Data Protection Act 2023 and the ABDM consent framework. The patient authenticates via ABHA ID, grants explicit consent for data capture and sharing, and the structured history is pushed to the hospital HIS/EMR and linked to the ABHA Personal Health Record via FHIR APIs.

- **Secure processing:** voice and document AI are processed securely within the software platform.
- **Session termination:** temporary session data is cleared immediately after submission.
- **Consent-first design:** granular, revocable consent with audio explanation for low-literacy patients.

3.4 End-to-End Patient Journey (as specified in the PS)

Step 1 — Identify: Patient logs into the software platform, enters/scans ABHA ID or Aadhaar details or registers as new; selects language; grants consent (audio-guided).

Step 2 — Converse: AI conducts adaptive voice + touch history interview, capturing chief complaint, HPI, and full history; red flags trigger priority triage.

Step 3 — Scan: Patient uploads prior prescriptions, lab reports, and discharge summaries; AI digitizes, structures, and timelines them.

Step 4 — Summarize & Route: AI generates structured history summary, links to ABHA, pushes to HIS, updates the patient's digital record; summary appears on physician's screen at consultation.

Step 5 — Consult: Physician reviews complete history in seconds, edits/confirms, and devotes the full consultation to examination, reasoning, and counselling.

PART 1 — The Core Issue (What Exactly Is Broken?)

1.1 The 2-to-5-Minute Consultation

History-taking is the single most important diagnostic activity in medicine — the evidence across five landmark studies (Lancet 1947 → BMJ 1975 → JAPI 2000 → UCLA 2009) shows the medical history alone yields the final diagnosis in **56–88% of cases, pooled mean 75.9%** — corroborated by modern literature (JMIR 2024: ~80%; Nature 2025: 60–80%) — before any examination or investigation. Yet in India's public hospital OPDs:

- Tertiary government hospitals register **4,000–10,000 OPD patients per day** — AIIMS New Delhi alone serves **~8,000–10,000 OPD patients daily (~35 lakh/year)** with only ~2,500–4,050 seats for waiting patients (Union Health Ministry figures via *The Hindu*, 14 Jan 2019; AIIMS charter; Hindustan Times, 9 Jul 2015). The PS sponsor itself, **AIIA, logged 3,28,193 OPD patients in FY 2024–25 alone** (Ministry of Ayush “11 Years of AYUSH” report, 16 Sep 2025).
- The **BMJ Open (2017)** systematic review across **67 countries, 179 studies, 28.5 million consultations** placed India's average primary-care consultation at **~2 minutes** — among the shortest globally (48 s in Bangladesh to 22.5 min in Sweden; 18 countries covering ~50% of world population at ≤5 min).
- Inside those 2–5 minutes, a physician must simultaneously elicit history, examine the patient, read prior paper records, diagnose, counsel, and prescribe.

The inevitable result: systematic under-elicitation of history, missed comorbidities and allergies, repeated questioning across visits, diagnostic error, and physician burnout.

*Every claim in Part 1 is backed by verified data — see Appendix A: Verified Evidence Base for the full source list with publication dates. Headline numbers: diagnostic errors occur in **5–20% of physician–patient encounters (WHO, 13 Sep 2023)** and **~1 in 14 hospital patients suffers a harmful — 85% preventable — diagnostic error, with history-taking failures carrying 2.5× risk (BMJ Quality & Safety, 2 Oct 2024)**; Indian hospitals show a **34% median medication-error incidence concentrated at the prescribing stage (systematic review, 2025)**; **98.6% of patients still keep paper records and 27% have lost them (Indian J Med Res, 2024)**; **90+ crore ABHAs exist (PIB, 30 May 2026) but under 1.6 lakh facilities run ABDM software (PIB, 11 Feb 2025)**; and **68,584 hit-and-run accidents in 2024 (MoRTH)** produced thousands of “blank patients” for whom emergency access is life-critical.*

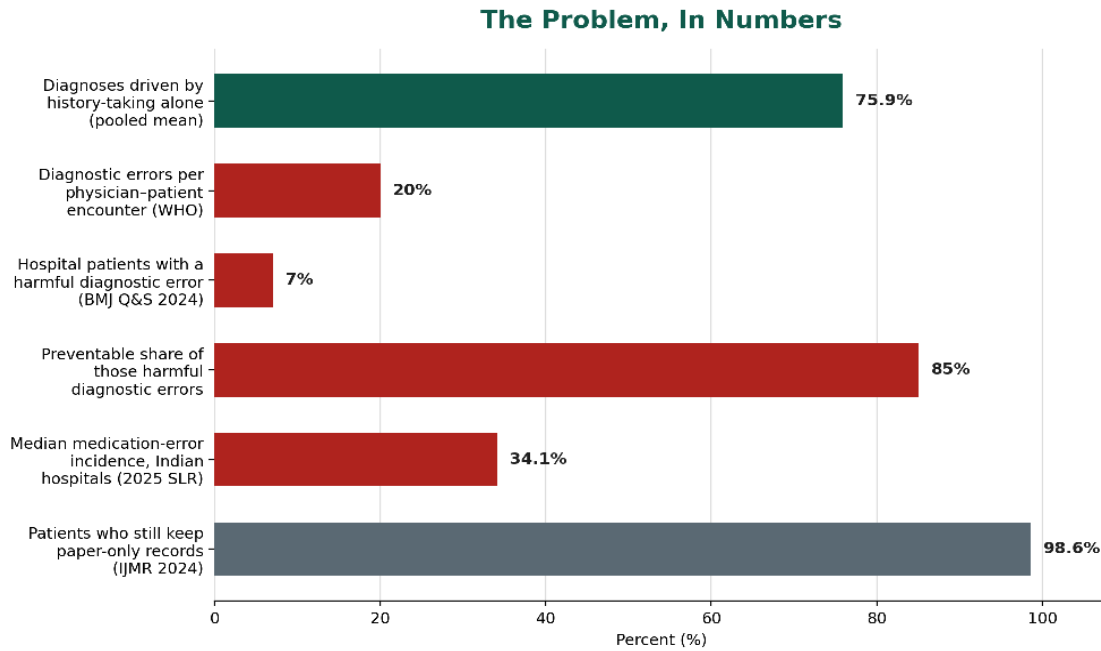


Figure 1. The scale of the problem, drawn from the verified evidence base (Appendix A).

1.2 The Paper Records Problem

Indian patients typically arrive carrying a plastic bag of paper — prescriptions, lab reports, discharge summaries, imaging films — from multiple previous providers. These documents are:

- **Unstructured** (free-form, half-filled templates)
- **Multilingual** (Hindi, English, regional languages, often mixed with Latin drug names)
- **Chronologically disordered** (the newest report is often at the bottom of the pile)
- **Frequently handwritten** and partially illegible

The physician burns a significant share of the consultation just *reading the bag*. There is no point-of-entry mechanism that digitizes, structures, and chronologically organizes these records before the patient reaches the consultation room.

1.3 The AYUSH Multiplier

AYUSH institutions (the PS sponsor is AIIA, Ministry of Ayush) face an extra layer: Ayurvedic case-taking requires **Trividha, Ashtavidha and Dashavidha Pariksha** — Prakriti (constitution), Vikriti (imbalance), Agni, Koshtha, Sara, Samhanana, Pramana, Satmya, Sattva, Ahara Shakti, Vyayama Shakti, Vaya, plus Ahara-Vihara (diet/lifestyle), Nidana and Samprapti analysis. This is a *far* deeper interview framework than allopathic intake. Capturing it manually inside a 2-minute OPD slot is effectively impossible — so practitioners abbreviate the very assessment that defines personalized Ayurvedic care.

1.4 The “First-Mile” Gap in ABDM

The Ayushman Bharat Digital Mission has built the rails — ABHA IDs, the Health Information Exchange, FHIR-based interoperability standards. But the **first mile is missing**: there is no patient-facing software that captures structured clinical history and digitizes documents *into* the ABDM ecosystem before the clinical encounter begins. Registration systems capture demographics; nothing captures the *clinical story*.

1.5 The Silent Failures

Beyond slowness, the current system has three dangerous failure modes:

Missed allergies / drug interactions — a penicillin allergy buried in a 3-year-old handwritten prescription never surfaces; a drug is prescribed that interacts with the patient's current medication.

Missed red flags — a patient with chest pain + breathlessness + sweating sits in the routine queue for 2 hours because nobody structured their complaint before triage.

The blank patient — in accidents and emergencies, the unresponsive patient arrives with *no history at all*. Doctors treat blind, with no access to prior diagnoses, medications, allergies, or even blood group.

PART 2 — Why It Matters

Stakeholder	What the status quo costs them
Patient	Repeats their story at every visit; waits hours; suffers missed allergies and delayed emergency care; loses paper records; gets generic (non-personalized) AYUSH care.
Doctor	Spends the majority of a 2–5 min consult on data-gathering instead of reasoning; faces medico-legal exposure from undocumented history; burnout from queue pressure.
Hospital	Low OPD throughput, duplicated tests (because prior reports can't be found), medico-legal risk, no analytics on case mix.
Ministry / Public Health	ABDM rails exist but carry no clinical data; no population-level insight from OPDs; AYUSH personalization remains theoretical.

A well-conducted history drives 70–80% of diagnoses. **Every minute a doctor spends reading a paper bag is a minute not spent thinking.** MediKiosk attacks exactly this waste.

PART 3 — The Solution: MediKiosk

3.1 One-Line Value Proposition

MediKiosk converts a patient's spoken words, touch responses, and paper medical records into a verified, structured, physician-ready clinical summary — before the consultation begins — and makes that history available in emergencies through consent-governed ABHA-based access.

3.2 What MediKiosk Is (and Is Not)

It is:

- A **patient-facing AI clinical intake platform** with two surfaces: a **walk-in kiosk/tablet** at the hospital and a **patient mobile/web app** for pre-visit and at-home intake.
- A **document digitization and intelligence pipeline** (OCR → classification → medical entity extraction → chronological timeline).
- A **physician dashboard** that renders the AI-drafted summary, the timeline, the original scans, and source/confidence labels — with full edit/verify/reject control.
- A **consent-first, ABDM-compatible integration layer** (ABHA, FHIR, HIS/EMR).
- An **emergency access (“break-glass”) service** letting verified healthcare organizations retrieve a critically needed history via ABHA ID under strict audit.
- A **tamper-evident record layer** using cryptographic hash-anchoring, so any alteration of a stored record is mathematically detectable.

It is NOT:

- A diagnostic engine. It never diagnoses, never prescribes, never replaces physician judgment. The AI output is always a *draft for a human to verify*.
- A blockchain gimmick. We deliberately use **lightweight hash-anchoring** (Merkle roots anchored on a permissioned ledger / signed WORM audit logs) rather than storing medical data on-chain — this preserves DPDP 2023 data-erasure rights while still guaranteeing tamper-evidence.

3.3 Design Philosophy — The Five Pillars

Speak first, type less — voice-first, touch-always. Every question is answerable by speaking *or* tapping; nothing requires literacy.

Consent before capture — granular, revocable, audio-explained consent, recorded with timestamp, purpose and version.

Human-in-command — every AI-extracted field carries a source label (`patient-reported`, `OCR-extracted`, `patient-verified`, `physician-verified`) and a confidence score. Nothing enters the permanent record unverified by a professional.

Fit the rails that exist — ABDM (ABHA, consent manager, FHIR), hospital HIS/EMR, DPDP 2023. We plug into the ecosystem; we don't ask hospitals to rip anything out.

Safety before speed — a rule-based red-flag engine (not probabilistic AI) can interrupt the routine flow and summon a human in seconds.

PART 4 — How It Works (Detailed)

4.1 The Two Windows (Dual-Surface Model)

MediKiosk provides **two coordinated windows**:

Window 1 — Patient (Kiosk / Mobile / Web App)

- **Identify:** ABHA QR scan / ABHA number / hospital ID / token number / new-patient flow. Language selection (Hindi, English + regional via Bhashini).
- **Consent:** audio-guided, granular consent (history capture / document processing / doctor sharing / ABDM linking) — each separately toggleable.
- **Quick Profile:** blood group, allergies, current medications, age, emergency contacts — captured once, editable forever.
- **Converse:** the adaptive AI history interview (voice + touch).
- **Scan:** upload/photograph prior prescriptions, lab reports, discharge summaries.
- **Chat:** a conversational assistant grounded in *their own* records (medication reminders, “what did my last report say?”, pre-visit guidance) with an Ayurveda mode.
- **Control:** view timeline, manage consent grants, see exactly which organizations accessed which record and when.

Window 2 — Verified Healthcare Organization (Web Portal)

- **Doctor dashboard:** queue, structured summary, timeline, original scans beside extracted text, red-flag alerts, edit/verify/reject/save-to-HIS.
- **Triage dashboard:** real-time red-flag alerts with patient location and token.
- **Emergency access (“break-glass”):** retrieve a critically needed record by ABHA ID under emergency protocol (see §4.6).
- **Admin analytics:** throughput, intake time, OCR correction rates, language usage — all anonymized.

Why two windows matters: the patient owns the data and the consent; the organization gets the clinical value. This mirrors ABDM's consent-manager architecture rather than fighting it.

4.2 End-to-End Patient Journey (Happy Path)

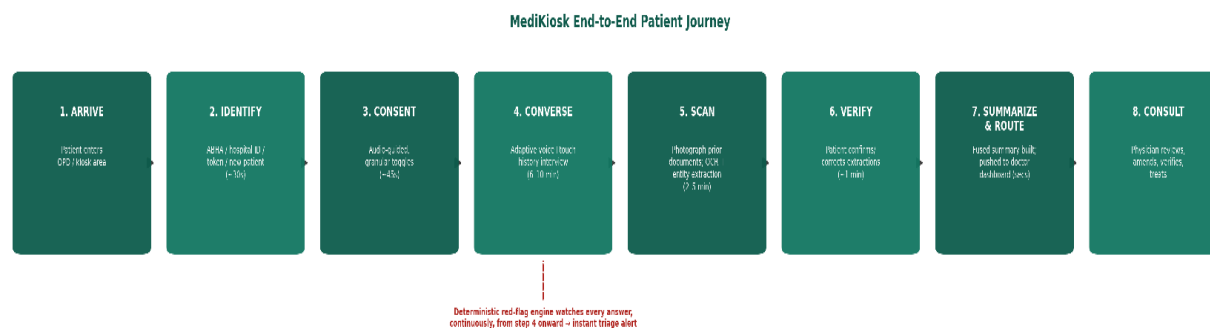


Figure 2. The MediKiosk end-to-end patient journey, from arrival to consultation.

Identify (≈30 s): Patient selects language, scans ABHA card or enters ABHA number / hospital ID / token; new patients create a temporary profile. Aadhaar is *not* the default medical identifier — ABHA/hospital ID/token are prioritized.

Consent (≈45 s): Audio explanation in the chosen language; separate toggles for history capture, document processing, doctor sharing, and optional ABDM linking; decline → graceful fallback to manual intake.

Converse (6–10 min): The conversational engine conducts the structured interview — chief complaint → adaptive follow-ups (SOCRATES/OPQRST) → past medical/surgical → drugs & allergies → family/personal → review of systems → optional AYUSH mode (Dashavidha Pariksha). Every question is voice-answerable *and* tap-answerable. Progress is visible. Answers can be corrected at any time.

Red-flag check (continuous): A deterministic rule engine watches every answer. “Chest pain” + “breathlessness” + “sweating” → immediate audio+visual message to patient, instant alert to triage dashboard, priority token. The routine interview may continue (or pause) under triage direction.

Scan (2–5 min): Patient photographs each paper document. Quality check on-device (blur/glare/crop) → auto re-capture prompts → document classification (prescription / lab report / discharge summary / referral / imaging / other) → OCR (printed + handwritten, multilingual) → medical entity extraction (diagnoses, medications + dosage/frequency/duration, investigation values + units + reference ranges, procedures, dates, issuing doctor/hospital).

Verify (≈1 min): The system reads back the extracted items; the patient confirms or corrects each (touch or voice). Low-confidence extractions are explicitly flagged for confirmation.

Summarize & Route (seconds): The summary engine fuses conversational history + verified document data into the standard physician format, builds the chronological timeline, and pushes to the doctor dashboard. On physician approval, the record maps to FHIR resources (Patient, Encounter, Condition, Observation, MedicationStatement, AllergyIntolerance, DocumentReference, Consent) and links to the ABHA PHR via ABDM consent APIs.

Consult: The doctor opens a completed, structured, source-labelled history — reads it in **seconds**, amends anything, verifies, and spends the consultation on examination, reasoning and counselling.

4.3 Module A — Conversational Multimodal History Engine

- **Dual-mode input:** every single question answerable by speaking OR tapping (icons, yes/no, body-map, sliders, multi-select). Literacy is never a gate.
- **Adaptive questioning:** a dialogue manager constrained by a clinician-approved clinical-history ontology branches on the chief complaint and prior answers. Saying “chest pain” triggers SOCRATES (Site, Onset, Character, Radiation, Associated symptoms, Timing, Exacerbating/relieving, Severity); “fever” triggers onset/duration/chills/rash/cough triads, etc. Question paths are **clinician-approved templates**, not free-running LLM dialogue — this is a safety feature.
- **Multilingual ASR/TTS:** Bhashini / AI4Bharat models for Indian languages and accents, with noise reduction tuned for hospital environments.
- **Completeness tracking:** a coverage meter per history section so the engine knows what is missing and asks follow-ups before closing.
- **AYUSH Mode:** extended interview capturing Dashavidha Pariksha (Prakriti, Vikriti, Sara, Samhanana, Pramana, Satmya, Sattva, Ahara Shakti, Vyayama Shakti, Vaya) plus Agni, Koshtha, Ahara-Vihara, Nidana and Samprapti — questionnaire-configurable per department.
- **Patient review step:** before submission, key answers are read back in local language for confirmation.

4.4 Module B — Medical Document Digitization & Intelligence

Pipeline per document: **capture** → **quality check** → **classification** → **OCR** → **layout analysis** → **entity extraction** → **confidence scoring** → **patient verification** → **timeline placement**.

- **Intelligent extraction:** diagnoses, medications (name/dose/frequency/duration), investigation results with values/units/reference ranges, procedures/surgeries, allergies, follow-up instructions, issuing provider, date.
- **Chronological organization:** documents auto-dated and merged into a coherent medical timeline (illnesses, admissions, surgeries, medication changes).
- **Abnormal-value highlighting:** out-of-range lab values and potential drug–drug or drug–allergy conflicts flagged for the physician.
- **Honest limitation handling:** every extracted item is labelled `extracted` / `patient-verified` / `physician-verified` with a confidence score. OCR output is *never* silently treated as clinical fact.

4.5 Module C — Structured Summary Generator + Physician Dashboard

Standard clinical format: Chief Complaint → HPI → Past Medical/Surgical → Drug & Allergy → Family → Personal/Social → ROS → Prior Investigations → Current Medications → Red-Flag Alerts → **Unverified Fields** (explicitly listed).

- **Editable & verifiable:** the summary is a draft to accept, amend, or reject — never an autonomous diagnosis. Physician controls: edit, verify, reject field, add note, request more history from patient, save-to-HIS.

- **Bilingual output:** patient-facing audio confirmation in local language; physician-facing summary in English/Hindi.
- **Source-linked:** every field shows its provenance (voice answer on 2026-09-04 / OCR of prescription dated 2025-11-15) and confidence.

4.6 Module D — Emergency Access (“Break-Glass”)

The differentiator no registration system has.

Scenario: an accident victim arrives unconscious. No bag of papers, no relative, no history. Today the team treats blind.

MediKiosk flow:

Verified hospital staff (authenticated, HFR-registered facility) enters/scans the patient's ABHA ID or Aadhaar-linked ABHA.

Emergency protocol triggers: minimal-verification retrieval (OTP to registered contact / Aadhaar demographic match where available), explicitly marked as emergency access.

A **restricted emergency view** opens: blood group, allergies, current medications, comorbidities (diabetes, epilepsy, cardiac history, anticoagulants), recent surgeries, implants — exactly what an ER team needs in the first 10 minutes.

Mandatory governance: every break-glass access is (a) written to an immutable audit log, (b) post-hoc reviewed by the hospital's privacy officer, (c) notified to the patient (SMS/app on recovery), (d) reported on the patient's access log. Abuse = revocation of the facility's verification.

This directly answers the PS's unspoken nightmare case and the Ministry's interest in ABDM actually *serving* emergency care.

4.7 Module E — Record Assistant (Patient Chat, Grounded + Ayurveda Mode)

- **Grounded RAG chat:** patients ask questions in text or voice; answers are retrieved *from their own verified records* (last HbA1c value, current medicines and timings, what a report term means in simple language, when the next dose is due). Guardrails: no diagnosis, no prescription advice — escalation to “consult your doctor / book appointment” for anything clinical.
- **Ayurveda Mode:** constitution-aware lifestyle guidance (diet, dinacharya, sleep) consistent with the patient's Prakriti/Vikriti captured in AYUSH Mode — educational content, clearly labelled as such.
- **Critical-condition pathway:** if the chat or interview surfaces red-flag patterns, the UI pivots to a one-tap “book emergency appointment / alert triage” action.

4.8 Module F — Consent, Privacy, Security & Tamper-Evidence

- **DPDP 2023 compliant:** purpose-limited collection, granular consent, right to correction, right to erasure, grievance contact. Data minimization is enforced at the questionnaire level (skip non-mandatory questions freely).

- **ABDM consent framework:** consent artefacts managed per ABDM spec; ABHA-linked sharing only with explicit consent; revocable at any time.
- **Security:** encryption in transit (TLS 1.3) and at rest (AES-256), role-based access control, session timeouts, device locking, temporary audio deleted after processing, full access audit trails, network segmentation, secure API gateway.
- **Tamper-evidence via hash-anchoring (no heavy blockchain):** medical documents and record payloads are stored encrypted off-chain (object storage + PostgreSQL), never on-chain — preserving erasure rights. On each record finalization, a SHA-256 hash (and periodic Merkle root aggregating many records) is anchored to a permissioned, append-only ledger (e.g., Hyperledger Fabric channel among verified healthcare organizations, or a signed WORM audit-log service). Any subsequent alteration of a record changes its hash → mismatch with the anchor → mathematically detectable tampering, with the audit log naming who touched what and when.
- **Session hygiene:** temporary session data cleared immediately after submission; retention policy applied per record class.

4.9 Data Classification & Handling

Data class	Examples	Protection
Identity data	Name, age, ABHA ID	High (RBAC, encrypted)
Health data	Symptoms, history, timeline	Very high (+ consent-gated)
Voice recordings	Raw ASR audio	Very high (deleted post-processing)
Documents	Scans, OCR text	Very high (+ hash-anchored)
Operational	Token, department	Medium
Anonymized analytics	Completion rates	Low

Separation principle: patient identity data lives in a normal relational database; medical content lives in an encrypted clinical store; the two are linked by an opaque internal identifier. An organization that legitimately holds the medical record cannot silently enumerate identities, and vice versa.

4.10 System Architecture (Summary)

MediKiosk — System Architecture



Figure 3. MediKiosk layered system architecture, from client surfaces to the ABDM/HIS integration layer.

Recommended stack: React/Next.js + Flutter (patient surfaces), React (doctor portal), FastAPI or Node.js backend, PostgreSQL + object storage (Supabase-compatible — matches team's existing codebase), Bhashini ASR/TTS, PaddleOCR + medical-layout models, constrained LLM for summarization behind a rule-based safety layer, FHIR server (HAPI), OAuth2/OIDC, deployment on hospital private cloud / MeghRaj (GI Cloud).

PART 5 — How Our Solution Maps to (and Enhances) the PS

PS requirement	MediKiosk delivery
Patient-facing, purpose-built software	Kiosk + app, zero-training UI, voice-first
Natural spoken conversation + guided touchscreen	Dual-mode engine on every question
Multilingual, multi-accent in noisy environments	Bhashini/AI4Bharat ASR + noise reduction
Low-literacy & elderly accessibility	Icon UI, audio prompts, assisted mode, large text/high contrast
Structured history (CC → HPI → ... → ROS)	Ontology-constrained adaptive interview, SOCRATES/OPQRST templates
AYUSH Dashavidha Pariksha	Dedicated AYUSH Mode + Ayurveda chat mode
Document OCR (handwritten+printed, multilingual)	Full digitization pipeline + entity extraction + timeline
Physician-ready structured summary	Module C, editable/verifiable, bilingual
Red-flag detection	Deterministic rule engine + triage alert + priority queue
DPDP 2023 + ABDM consent compliance	Consent-first design, granular, revocable, audited
HIS + ABHA integration	FHIR-mapped, consent-manager compliant
(Enhancement) Emergency access via ABHA	Break-glass module with governance
(Enhancement) Tamper-proof records	Hash-anchoring on permissioned ledger
(Enhancement) Patient chat grounded on records	RAG assistant + Ayurveda mode

PART 6 — Comparison With Existing Solutions

Solution class	What it does	Why it falls short vs. MediKiosk
Hospital registration systems (current HIS registration modules)	Demographics, appointment, token	Zero clinical history, no document processing, no ABHA clinical linkage
Patient check-in kiosks (developed-country hospitals)	Administrative check-in only	No AI history acquisition, no OCR, no multilingual voice, not built for 2-minute OPD realities
Mobile health apps / tele-triage chatbots	Symptom checkers, bookings	Need smartphone literacy, connectivity, pre-enrolment — excludes the elderly, rural, low-literacy, first-visit majority of government OPDs
Nurse-led triage / history desks	Human interview before consult	HR-limited, doesn't scale to 5,000+/day, reintroduces the transcription bottleneck
Generic document scanners / DMS	Digitize images	No OCR intelligence, no entity extraction, no chronology, no link to history or ABHA
ABDM ecosystem apps (ABHA creation, PHR apps)	Identity + record linking	Assume the records already exist digitally; they don't solve creation of structured history from paper and speech
EMR/EHR systems	Doctor-facing documentation	Shift the typing burden to the doctor — worsen the 2-minute problem; none are patient-facing intake
MediKiosk	All of the above gaps in one platform: voice+touch intake, document AI, red flags, AYUSH mode, physician verification, ABDM/HIS integration, emergency access, tamper-evidence	—

The honest framing for judges: we are not competing with ABDM — we are the missing first-mile producer of structured clinical data that makes ABDM's rails carry real payloads. We are not competing with HIS — we feed verified summaries into HIS.

PART 7 — Uniqueness, Innovation & USP

Innovation claims (defensible)

First patient-facing conversational clinical-intake platform designed for Indian public OPD constraints — 2-minute-consultation reality, multilingual, low-literacy, zero-training.

Dual-mode (voice ⊕ touch) on every question — most “voice AI” demos fail the literacy test; we treat touch as a first-class equal, not a fallback.

Ontology-constrained adaptive dialogue — LLM fluency caged inside clinician-approved question paths; adaptive like a doctor, safe like a protocol.

Fused intake: conversational history + OCR'd paper records merge into one verified timeline — nobody else combines both at point of entry.

Deterministic red-flag safety layer — emergency escalation that doesn't depend on probabilistic AI confidence.

AYUSH-native design (the PS sponsor's own need) — Dashavidha Pariksha as a configurable questionnaire mode; allopathic systems don't even attempt this.

Break-glass emergency access via ABHA with governance — turns ABDM from a filing system into a life-saving emergency tool.

Hash-anchored tamper-evidence without blockchain baggage — integrity guarantees compatible with DPDP erasure rights.

Provenance/confidence labelling on every data field — an audit trail of knowledge, not just access.

USP (the one sentence for the pitch): “MediKiosk is the only platform that turns what a patient says and carries on paper into a doctor-verified, emergency-accessible, AYUSH-aware digital history — in their own language, before the doctor calls their token.”

PART 8 — Feasibility

8.1 Technical

- **ASR for Indian languages:** mature via Bhashini (Govt. of India National Language Translation Mission) and AI4Bharat open models; noisy-environment robustness is engineering, not research.
- **OCR (printed + handwritten, multilingual):** production-grade engines exist (PaddleOCR, cloud OCR, Indic-specific models); handwriting remains the hardest nut — mitigated by patient-verification step and honest confidence labels.
- **Constrained clinical dialogue:** LLM + clinician-approved ontology/templates = demonstrable today; no exotic research needed.
- **ABDM/FHIR:** public sandboxes and well-documented APIs (HAPI FHIR); mock-first integration strategy de-risks the demo.
- **Team readiness:** existing React/TypeScript/Vite/Supabase codebase (MediCareX, Clinic Ease) provides the frontend + backend scaffolding; Agents.md already specifies modules, data objects, and demo scenarios.

8.2 Operational

- Kiosk hardware = commodity tablet + mic/speaker/camera; assisted mode covers device failures; offline queueing handles network drops.
- Staff role is supervisory (help, triage alerts), not transcriptional — net reduces HR load.
- Migration is phased: standalone pilot → registration integration → HIS/EMR push → FHIR/ABDM. No big-bang replacement.

8.3 Regulatory / Legal

- DPDP 2023: consent-first, minimization, erasure-compatible architecture (no on-chain data).
- ABDM: consent-manager artefacts, purpose-limited sharing.
- Clinical governance: system never diagnoses/prescribes; physician verification is mandatory before HIS write; red-flag rules clinician-approved.
- Data localization: deployable on hospital private cloud / MeghRaj.

8.4 Feasibility Verdict

High. Every component relies on mature or government-backed (Bhashini, ABDM sandbox) technology; the innovation is in the integration and workflow design, which is exactly what a 36-hour hackathon prototype can demonstrate credibly.

PART 9 — Scalability & Extensibility

- **Horizontal scale:** stateless API services scale behind a gateway; AI calls are the bottleneck → queue-based async processing, GPU autoscaling, and on-device ASR caching where possible. One kiosk handles ~40–60 intakes/day; a 10-kiosk hospital processes 400–600 pre-structured patients/day.
- **Multi-hospital:** tenant-per-hospital data partitioning; shared AI services; central anonymized analytics.
- **Configurability:** questionnaires are data (JSON templates), not code — new complaints, specialties, languages, and AYUSH protocols deploy without engineering.
- **Language scale:** architecture is language-agnostic via Bhashini; adding a language = adding templates + TTS/ASR model, not re-architecture.
- **Roadmap:** Phase 1 prototype (Hindi/English, 5 complaint pathways, printed OCR, mock ABDM) → Phase 2 pilot (more languages, handwriting, AYUSH module, HIS integration, offline) → Phase 3 production (full ABDM/FHIR, multi-hospital, device management) → Phase 4 advanced (follow-up comparison, medication-adherence questions, patient education audio, specialist modules, mental-health screening with safeguards, anonymized population analytics).
- **The compounding effect:** every visit enriches the longitudinal timeline — follow-up visits become faster than first visits, unlike every paper-based system where each visit restarts from zero.

PART 10 — Adaptability: Who Uses It and Why

User	Why they use it (despite existing options)
Elderly / low-literacy rural patient	Speaks instead of types; hears questions in own language; nobody asks them to read or write. No app they currently have does this.
First-time / walk-in OPD patient	No pre-enrolment needed; ABHA created/linked on the spot; token + summary ready before consultation.
Government OPD doctor	Walks in to a finished, structured, source-labelled history; edits 30 seconds instead of eliciting 8 minutes; red flags already escalated; medico-legal documentation improved.
AYUSH practitioner (AIIA etc.)	Finally gets Dashavidha Pariksha captured in full without doubling consult time — the PS sponsor's own pain point.
Triage / nursing staff	Gets deterministic emergency alerts with location and reason instead of discovering crises by chance.
ER team (accident cases)	Break-glass access to allergies/medications/comorbidities of an unconscious stranger — today this information simply does not exist at point of care.
Hospital administration	OPD throughput up, duplicated investigations down, ABDM-compliance demonstrable, audit-ready records.
Ministry of Ayush / public health	Structured AYUSH intake data at population scale; ABDM rails finally carrying clinical payloads.

Why adoption beats the alternatives: existing options each solve one slice (registration, or scheduling, or document storage). MediKiosk is the only option whose unit of value is the physician's scarcest resource — consultation minutes — and whose failure mode is a graceful fallback to today's manual process (manual intake always remains possible).

PART 11 — Impact (With Measurable Targets)

Dimension	Impact mechanism	Pilot target (presented as goals)
Doctor time	History pre-elicited and pre-structured	30–50% reduction in history-taking time; consult reallocated to examination/counselling
Patient wait	Parallel intake while queueing	Registration-to-consultation time reduced materially; intake completed in 8–12 min during the wait patients already endure
Safety	Red-flag rules + allergy/drug-interaction surfacing	100% of configured red-flag patterns routed to triage; allergy list present on every summary
Completeness	Ontology-driven coverage meter	>80% question completion; near-elimination of repeated questioning across visits
Records	OCR + timeline	Prior paper records chronologically organized for 100% of patients who carry them
AYUSH care quality	Full Dashavidha capture	Personalized (Prakriti-aware) care becomes operationally feasible in OPD time
Emergencies	Break-glass access	Critical history available in minutes for unidentified/unresponsive patients
Systemic	ABDM data creation	First-mile structured data flowing into national digital-health rails; reduced duplicate investigations

PART 12 — MVP Scope for SIH (What We Build in 36 Hours)

Patient side

Hindi + English; new/existing patient flow; consent screen; voice+touch interview for **fever** and **chest pain** pathways (demonstrates adaptivity *and* red-flag); document image upload; OCR for printed lab report/prescription; basic entity extraction; patient confirmation screen; structured summary generation; mock ABHA/HIS integration.

Doctor side

Login; patient queue; structured summary with sections; original document preview beside extracted text; red-flag banner; edit/verify/reject; final save.

Demo story (single take)

Hindi-speaking patient arrives with chest discomfort + an old lab report → selects Hindi → scans ABHA → audio consent → says “seene mein dard hai” → adaptive SOCRATES questions → reports sweating + breathlessness → **red-flag alert fires, triage dashboard pings** → scans lab report → OCR extracts sugar/cholesterol values → patient verifies → summary appears on doctor dashboard → doctor corrects one field, verifies, saves → record maps to FHIR and (mock) pushes to ABDM. **Second pass:** AYUSH Mode + emergency break-glass retrieval on a tablet. Q&A-ready.

Explicit non-goals for MVP (state these proudly)

No autonomous diagnosis, no autonomous prescription, no claim of perfect handwriting OCR, no real HIS connection (mocked), 2 languages only, synthetic data only.

PART 13 — Risks & Mitigations

Risk	Mitigation
Handwritten OCR errors	Patient-verification step + confidence labels + physician verification gate
ASR failure in noise/certain accents	Touch fallback on every question; assisted mode; per-language model selection
False red-flag alerts	Deterministic rules tuned with clinicians; triage human decides; false-alert rate tracked
Patient distrust of AI/consent fatigue	Audio explanations, granular toggles, visible data deletion, staff assistance
Hospital IT integration resistance	Phased adapters; standalone-pilot first; FHIR standards; no rip-and-replace
DPDP non-compliance	No on-chain data; erasure-compatible stores; privacy-officer review of break-glass
LLM hallucination in summaries	Constrained generation to structured fields; rule-based safety layer; all output physician-verified
Device vandalism/downtime	Rugged enclosures; offline queueing; manual-intake fallback always available

PART 14 — FAQs (Anticipated Judge/Mentor Questions)

Q1. Isn't this just a chatbot asking questions?

No. A chatbot does free dialogue. MediKiosk is an ontology-constrained clinical interview engine with dual-mode input, completeness tracking, deterministic red-flag safety, document-AI fusion, and a mandatory human-verification gate. The LLM is a component, not the product.

Q2. What if the patient can't read, can't speak clearly, or is too frail?

Touch/icons answer every question without literacy; staff-assisted mode exists; family members can complete intake with the patient present; assisted mode is a designed path, not an error state.

Q3. How accurate is the OCR, especially handwriting?

Printed documents: high accuracy with modern engines. Handwriting: assisted by patient verification — the patient confirms or corrects each extraction, and the physician verifies before the record is finalized. Every field carries a confidence score; nothing low-confidence is silently trusted.

Q4. Does the AI diagnose?

Never. It structures, summarizes, and flags emergencies by deterministic rules. Diagnosis and prescription remain exclusively with the physician, who must verify the summary before it enters the HIS.

Q5. Why did you drop blockchain? Your earlier pitch mentioned it.

Deliberately. On-chain medical data conflicts with DPDP 2023 erasure rights, adds latency/cost, and buys nothing a permissioned hash-anchor doesn't. We anchor SHA-256 hashes/Merkle roots of finalized records on an append-only permissioned ledger — tampering becomes mathematically detectable, while data stays erasable and fast. Integrity without the baggage.

Q6. How does the emergency access respect privacy?

Break-glass is restricted to authenticated staff of verified (HFR-registered) facilities, retrieves a minimal emergency view, is 100% audit-logged, post-hoc reviewed, and notified to the patient. Repeated unjustified use revokes facility access.

Q7. What about hospitals with no internet / power cuts?

Offline queueing with store-and-forward sync; kiosks can run on UPS; assisted manual intake is always the fallback. The design degrades gracefully instead of failing closed.

Q8. How does this integrate with our existing HIS?

Phased: standalone pilot → registration-system adapter → HIS/EMR push of doctor-approved summaries → FHIR/ABDM linking. We never require replacing existing systems.

Q9. What about dialects and code-mixed speech (Hinglish)?

Bhashini/AI4Bharat models support code-mixed Indian speech; the dialogue manager normalizes into structured fields; language can be switched mid-interview.

Q10. Who owns the data?

The patient. Consent is granular and revocable; every access appears in the patient's access log; ABDM linking is opt-in.

Q11. Can't a nurse just do this?

That's today's model — and it's exactly the bottleneck: it doesn't scale to 5,000+ daily patients, consumes the same scarce HR, and reintroduces transcription errors. MediKiosk makes staff supervisors of the process instead of its labor.

Q12. What stops a hospital from accessing records without consent?

RBAC + consent artefacts enforced at the API layer + hash-anchored audit logs + break-glass governance + patient-visible access logs. Unauthorized access is both technically blocked and forensically traceable.

Q13. Is this compliant with DPDP 2023 and ABDM?

Yes by design: purpose limitation, minimization, granular consent, correction/erasure rights, data localization options, and ABDM consent-manager artefacts.

Q14. What languages are supported at launch?

Hindi and English for MVP; architecture supports 10+ Indian languages via Bhashini; adding a language is configuration, not re-engineering.

Q15. What happens if the patient declines consent?

Graceful fallback: manual registration and traditional history-taking proceed exactly as today. Nothing is blocked.

Q16. How is this different from PHR apps like the ABDM-linked record apps?

PHR apps assume digital records already exist. MediKiosk creates structured digital history from speech and paper — the first mile those apps are missing.

Q17. Cost per hospital?

Commodity tablets/kiosks + cloud or on-prem deployment; the AI layer uses government-backed (Bhashini) and open-source components to keep marginal cost per intake near-zero. Pilots can run on as little as 2–4 tablets.

Q18. Why would a busy government doctor actually open this dashboard?

Because it arrives before the patient: the summary is on screen when the token is called, abnormalities and allergies are pre-highlighted, and editing is faster than eliciting. The 30-second review replaces 8 minutes of questioning.

PART 15 — Closing Narrative (For Pitch Decks)

India's doctors diagnose in two minutes not because the medicine is simple, but because the history arrives unwritten, scattered across a plastic bag of paper, in a language the system doesn't speak. MediKiosk gives every patient — literate or not, young or elderly, allopathic or Ayurvedic — a voice in their own language, turns their paper past into a verified digital timeline, puts a finished clinical summary on the doctor's screen before the token is called, summons help in seconds when seconds matter, and hands the emergency team the history of a patient who can't speak at all. We don't replace the doctor. We give the doctor back the one thing the system steals from them: time to think.

Appendix A — Verified Evidence Base (Problem Data With Sources)

Every statistic below is from a verified, citable source. Each entry lists the **data point**, the **source**, the **publication/access date**, and the **URL**. Authority tiers: peer-reviewed journals (S), government/PIB/MoRTH (S), major national media (A). Compiled: 4 September 2026.

1. The 2-Minute Consultation

India's average primary-care physician consultation length is **~2 minutes** — among the shortest in the world. The systematic review covered **111 publications, 179 studies, and 28,570,712 consultations across 67 countries**; global range: 48 seconds (Bangladesh) to 22.5 minutes (Sweden). **18 countries representing ~50% of the global population** have average consultations of **5 minutes or less**. Shorter consultations are significantly associated with more ambulatory-care-sensitive hospital admissions, lower physician satisfaction, and higher workload/stress.

Source: Hobbs FR et al., "International variations in primary care physician consultation time: a systematic review of 67 countries," BMJ Open 2017;7:e017902 (peer-reviewed, S). Published 8 November 2017. bmjopen.bmj.com/content/7/10/e017902 · pubmed.ncbi.nlm.nih.gov/29118053. Corroborating: The Times of India, 9 Nov 2017.

2. OPD Overload at Tertiary Hospitals

AIIMS New Delhi alone serves ~8,000–10,000 OPD patients per day (~35 lakh OPD patients/year), with only 2,500–4,050 waiting seats for them; its centralized registration centre runs **52 counters** built for 10,000 patients/day, and **2,250 registrations/hour** are required to keep up. Average attendance figures: 7,200/day (AIIMS official site), 8,000/day (Union Health Ministry figures reported by The Hindu), 10,000/day (Hindustan Times / PIB-reported).

Sources: The Hindu, "AIIMS to shift its OPD to more spacious accommodation," 14 Jan 2019 (citing Union Health Ministry). AIIMS New Delhi official website, Hospital Charter. Hindustan Times, 9 Jul 2015. TCS press release on AIIMS OPD transformation, 28 Dec 2015.

3. The PS Sponsor's Own Load (AIIA)

The **All India Institute of Ayurveda (AIIA)** — the sponsoring organisation of this problem statement — alone catered to **3,28,193 OPD patients and 8,317 IPD patients in FY 2024–25**, and distributed 10,000 Ayush Raksha kits + ran a health camp for 15,000 pilgrims at Maha Kumbh 2025.

Source: Ministry of Ayush, "11 Years of AYUSH" report, published 16 Sep 2025 (ayush.gov.in). Supporting: PIB, Ministry of Ayush Year-End Review 2025, 26 Dec 2025.

4. History-Taking Drives the Majority of Diagnoses

Across five landmark studies spanning 1947–2009, the medical history alone led to the final diagnosis in **56%–88%** of cases; pooled mean **75.9% (95% CI 67.9–83.9)**:

Study	History-alone diagnostic yield
Platt, The Lancet (1947)	74% (68–74/100)
Hampton et al., BMJ (1975)	82.5% (66/80)
Peterson et al. (1992, replicated Hampton)	76.3% (61/80)
Journal of the Association of Physicians of India (2000)	78.6% (77/98)
Wang et al., UCLA neuro-ophthalmology clinic (2009)	88% (of 115)

Modern corroboration: a 2024 JMIR study found medical history contributes **~80%** to a diagnosis, and a 2025 *Nature* paper (Google AMIE) notes **“60–80% of diagnoses are made through clinical history-taking alone.”**

Sources: Zander D., “A History of Patient History-Taking,” *Clinical Correlations* (NYU Grossman), 26 Jan 2022. JMIR Medical Education 2024 (mededu.jmir.org/2024/1/e52674). Tu T. et al., “Towards conversational diagnostic artificial intelligence,” *Nature* (2025).

Pitch line: “We are not automating the least valuable part of the visit — we are securing the most valuable part: the 75–80% of diagnostic information that today never gets captured.”

5. Diagnostic Error Is Common — and History-Taking Failures Are a Proven Cause

A. WHO

Diagnostic errors occur in **5–20% of physician–patient encounters**, and **“most people will suffer a diagnostic error in their lifetime.”**

Source: WHO Fact Sheet — Patient Safety, updated 13 Sep 2023.

B. BMJ Quality & Safety, 2024

Harmful diagnostic errors affect **~1 in every 14 (~07%) general-medicine hospital patients**; **85% were preventable**. Critically for this PS: errors were significantly associated with **history-taking failures (2.5× the risk)** and uncertainty in initial assessment (4× the risk).

Source: Rodziewicz TL / BMJ Group summary of the BMJ Quality & Safety single-centre study, published 2 Oct 2024.

C. US Multi-Hospital Study

In 29 hospitals, **23% of patients** who were transferred to ICU or died had experienced a diagnostic error; **17% of all patients suffered harm** from it.

Source: Schnipper/JAMA Internal Medicine study, reported by Harvard Gazette, 9 Jan 2024.

6. Medication Errors & Adverse Drug Events in India

A 2025 systematic literature review of **40 Indian studies** found a **median medication-error incidence of 34.11%** and error frequency of **26.74%** in Indian hospitals; errors concentrate at the **prescribing stage (40%)**

— the exact stage where missing allergy/drug-history information does its damage. India is estimated to see **~5.2 million injuries annually** from medical errors.

Source: Govil D. et al., “Epidemiology of Medication Errors in Indian Hospital Settings,” systematic review, *PMC*, 2025 (covering Jan 2014–Apr 2025).

Supporting (LMIC tertiary-care study): **36.2% of adverse drug events were medication errors; 88% occurred at the prescribing stage; 62% of all ADEs were preventable** — the dominant patient-side risk factor being polypharmacy without a consolidated medication list (≥ 7 medications → 3.5× odds of an ADE).

Source: Sahilu T. et al., “Adverse Drug Events and Contributing Factors Among Hospitalized Patients,” *PMC*, 2020.

Pitch line: “A consolidated, always-present allergy and medication list is one of the cheapest patient-safety interventions that exists — and it is exactly what a 2-minute consult cannot produce.”

7. Paper Records & Record Loss

In a peer-reviewed study of urban residents in northern India (n=419): **413 of 419 respondents (98.6%) maintained paper-based records; 96 of 351 (27.4%) had lost medical records; 57 (16.2%) had forgotten to bring records** to a healthcare visit. Separately, EMR adoption in India is estimated at only **~35% of hospitals**, concentrated in large urban private facilities; government hospitals, small clinics and rural centres remain predominantly paper-based (vs ~96% hospital EMR adoption in the US).

Source: “Perception towards electronic health records & uptake of digital health IDs among urban residents in northern India,” *Indian Journal of Medical Research*, 2024. Supporting: EMR adoption comparison, *logicloom.in*, Oct 2024.

8. ABDM — The Rails Exist, the First Mile Doesn't

A. ABHA Growth

ABHA accounts crossed 90 crore in 2026 — growth: 14.7 Cr (2021) → 30.4 Cr (2022) → 50.6 Cr (2023) → 72.2 Cr (2024) → 84.5 Cr (2025) → **90+ Cr (May 2026)**; women are ~half of all holders.

Source: PIB, “Ayushman Bharat Digital Mission Crosses Landmark Milestone of 90 Crore ABHA Accounts,” 30 May 2026.

B. The Asymmetry

As of 5 August 2025: 79.91 Cr ABHAs, 4.18 lakh facilities on HFR, 6.79 lakh professionals on HPR — but only **1.59 lakh health facilities actually use ABDM-enabled software** (as of Feb 2025), i.e., well under half of registered facilities generate digital encounters, and the vast majority of these records enter the ecosystem already digitized by the provider — not captured from the patient's speech and paper at the point of care.

Source: PIB, Rajya Sabha reply, “Update on ABDM,” 12 Aug 2025; PIB, 11 Feb 2025.

C. Scale of the Government OPD Universe

1,75,560 Ayushman Arogya Mandirs operational with **371.97 crore cumulative footfall** (till 31 Dec 2024); **31.86 crore eSanjeevani teleconsultations** delivered by Dec 2024; **36.36 crore Ayushman Cards** created under PM-JAY with 8.59 crore hospital admissions.

Source: Department of Health & Family Welfare, MoHFW, Annual Report 2024–25, published September 2025.

Pitch line: “90 crore Indians have a digital health identity. Almost none of their OPD histories are digital. We are the first-mile producer ABDM is missing.”

9. The Emergency “Blank Patient”

In **2024 alone, India recorded 68,584 hit-and-run road accidents killing 34,030 victims and injuring 51,517** — every one of them, when survivable, is a patient whose identity, allergies, medications and comorbidities are unknown to the treating team. Total road deaths in 2024: **~1.77 lakh** (1,51,950 male + 25,225 female), with thousands of victims of unknown age/identity.

Source: Ministry of Road Transport & Highways (MoRTH), “Road Accidents in India 2024,” official annual report, 2025.

Pitch line: “Every day, ~94 Indians survive a hit-and-run and reach a hospital as a stranger. Today that team guesses. With MediKiosk break-glass, it retrieves a medically critical summary in minutes — under full audit.”

10. Physician Workforce Distribution

India had **10,22,859 registered allopathic doctors as on 31 March 2017**, giving an allopathic doctor-to-population ratio of just **0.77 per 1,000** (per Lok Sabha answer reported in a peer-reviewed analysis) — and doctors are heavily concentrated in urban areas, which is why a single AIIMS OPD absorbs 10,000 patients/day while rural areas face scarcity. (Counting AYUSH practitioners, the combined ratio crosses 1:1,000 — which is precisely why this PS expects one platform to serve both systems.)

Source: Kumar R., “India achieves WHO recommended doctor population ratio,” Indian Journal of Health Sciences & Biomedical Research, PMC, 2018.

How to Use This Evidence in the Pitch

Open with the sponsor: AIIA alone did 3.28 lakh OPD consultations in one year (Ayush report, Sep 2025) — “this problem lives in your own OPDs.”

Quantify the waste: 2-minute consults (BMJ Open 2017) × 8,000–10,000 daily OPD load (AIIMS/Health Ministry) = the history that drives 76–80% of diagnoses (Hampton 1975 / JAPI 2000) is the first casualty.

Quantify the danger: 5–20% of encounters carry diagnostic error (WHO 2023); history-taking failure multiplies that risk 2.5× (BMJ Q&S 2024); 34% median medication-error incidence in Indian hospitals (2025 SLR) with errors concentrated at prescribing.

Quantify the readiness: 90 crore ABHAs (PIB, May 2026) but <1.6 lakh facilities on ABDM software (PIB, Feb 2025) — identity is solved, capture is not.

Quantify the emergency need: 68,584 hit-and-run accidents in 2024 (MoRTH) = thousands of “blank patients” per year for whom break-glass access is the difference between guessing and knowing.

All sources accessed and verified on 4 September 2026.